

MERIMBULA, Australia

WMO: 959290

Lat: 36.9078S

Lon: 149.8989E

Elev: 7

StdP: 14.69

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.1	37.8	29.8	23.9	53.6	31.8	26.2	50.5	20.8	60.1	18.6	58.9	3.1	220	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.9	80.8	69.1	78.2	68.3	76.4	67.4	71.6	77.7	70.2	75.8	69.2	74.6	11.5	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.3	107.7	75.1	68.2	103.5	74.1	67.0	99.4	73.1	35.4	77.9	34.2	75.8	33.3	74.7	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	17.4	15.3	DB	32.6	91.8	1.2	5.0	31.7	95.4	31.0	98.4	30.3	101.2	29.4	104.9	(4)
(5)			WB	31.8	74.0	1.1	1.8	31.0	75.4	30.3	76.4	29.6	77.5	28.8	78.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.0	67.4	67.3	65.2	61.1	55.9	52.1	50.5	51.3	54.5	57.8	61.2	64.5	(6)	
	DBStd	7.03	3.91	3.68	3.74	3.46	3.22	3.02	2.89	3.33	4.02	4.45	4.17	4.28	(7)	
	HDD50	66	0	0	0	0	0	12	28	21	4	1	0	0	(8)	
	HDD65	2478	20	17	42	123	281	388	450	424	318	228	126	61	(9)	
	CDD50	3359	538	484	473	334	184	74	42	62	138	243	337	450	(10)	
	CDD65	298	94	81	50	7	0	0	0	0	1	6	13	46	(11)	
	CDH74	1151	355	260	163	38	5	0	0	2	29	65	55	179	(12)	
(13)	CDH80	155	44	16	14	7	0	0	0	0	6	21	11	36	(13)	
(14)	Wind	WSAvg	6.1	7.0	6.4	5.9	5.3	5.4	5.3	5.4	5.9	6.3	6.7	7.1	6.9	(14)
(15) Precipitation	PrecAvg	32.8	3.0	3.1	3.6	3.3	2.6	2.7	1.6	1.6	2.2	2.7	3.4	2.6	(15)	
	PrecMax	56.9	7.1	15.4	10.1	17.6	11.0	12.4	5.2	6.2	5.7	12.9	10.6	11.1	(16)	
	PrecMin	12.4	0.6	0.1	0.3	0.2	0.0	0.1	0.0	0.2	0.2	0.4	0.2	0.1	(17)	
	PrecStd	10.5	2.1	3.2	2.8	4.3	2.6	3.1	1.5	1.4	1.6	3.0	2.7	2.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.3	82.4	80.8	78.2	73.5	67.5	68.2	71.3	77.6	80.9	79.5	84.0	(19)	
		MCWB	70.9	72.2	70.5	63.4	60.2	54.9	53.3	54.5	58.2	61.5	66.3	69.0	(20)	
	2%	DB	80.1	78.9	77.5	73.7	69.7	64.4	64.3	66.4	70.9	73.6	74.7	78.0	(21)	
		MCWB	70.2	69.9	68.5	63.7	58.8	54.5	52.5	53.3	56.7	61.5	65.2	67.8	(22)	
	5%	DB	77.4	77.1	75.6	71.6	67.2	62.5	61.8	63.0	66.8	70.0	72.2	75.1	(23)	
		MCWB	68.9	68.7	67.3	63.0	57.6	53.6	51.7	52.7	56.6	60.8	63.8	66.9	(24)	
	10%	DB	75.5	75.3	73.5	69.6	64.7	60.5	59.7	61.0	64.2	67.3	70.1	73.0	(25)	
		MCWB	67.7	67.6	66.0	61.6	55.9	53.0	50.7	51.7	55.0	59.1	62.4	65.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.2	73.3	72.0	67.4	63.1	59.6	57.5	58.2	61.3	65.8	68.9	71.5	(27)	
		MCDB	81.4	79.6	78.1	74.7	68.3	63.0	62.7	65.3	69.2	72.8	75.2	79.3	(28)	
	2%	WB	71.4	71.6	70.1	65.7	61.0	56.9	55.1	56.1	59.5	63.8	66.9	69.4	(29)	
		MCDB	78.0	77.0	75.2	71.4	66.8	61.5	61.2	63.1	66.8	70.1	72.7	75.9	(30)	
	5%	WB	70.1	70.2	68.8	64.3	59.3	55.2	53.1	54.4	58.0	62.2	65.3	68.0	(31)	
		MCDB	75.9	75.3	73.9	69.6	65.3	59.9	59.5	60.7	64.9	68.4	70.7	73.8	(32)	
	10%	WB	69.0	68.9	67.4	62.8	57.4	53.8	51.7	52.8	56.4	60.6	63.7	66.6	(33)	
		MCDB	74.4	74.1	72.5	68.1	62.8	58.7	58.2	59.1	63.0	66.0	68.9	71.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	14.2	15.7	17.1	18.8	17.9	20.3	20.4	20.0	18.2	15.9	15.1	(35)	
		MCDBR	17.6	16.5	18.4	20.2	22.8	21.0	23.8	23.8	24.6	23.2	19.1	18.4	(36)	
	5% WB	MCWBR	10.4	9.6	10.8	11.7	13.5	12.6	13.8	13.6	13.5	12.9	11.1	10.6	(37)	
		MCWBR	16.0	14.7	16.1	17.5	20.2	17.1	21.1	21.4	21.9	19.8	16.8	16.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.422	0.384	0.351	0.331	0.315	0.303	0.293	0.305	0.334	0.347	0.369	0.376	(40)	
		taud	2.340	2.491	2.567	2.596	2.590	2.616	2.639	2.603	2.521	2.515	2.487	2.482	(41)	
	Ebn at Noon		290	294	292	279	266	260	272	285	293	302	304	305	(42)	
		Edh at Noon	42	35	30	26	22	20	21	25	31	34	36	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2073	1782	1508	1141	850	666	779	1046	1439	1775	1960	2122	(44)	
	RadStd		157	137	108	84	42	45	60	54	98	101	139	140	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+81

Nomenclature: See separate page